Sandra Trajkovski

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SUMMARY

Biomedical engineer with a background in biosignal processing, machine learning on clinical data, and teaching signal processing at the undergraduate level. Experience instructing and tutoring across the bioengineering curriculum, building hardware and test setups, and communicating technical results to clinical and engineering audiences. Currently a process engineer at Applied Medical.

EDUCATION

University of California, Berkeley Aug 2025 – May 2026
Master of Engineering in Bioengineering | Magna Cum Laude

California State University, Long Beach Aug 2021 – May 2025
Bachelor of Science in Biomedical Engineering | Summa Cum Laude | Engineering Honors Program

EXPERIENCE

Process Engineer Rancho Santa Margarita, CA
Applied Medical | Packaging Department Aug 2026 – Present

- Develop, troubleshoot, and improve medical device packaging processes in a vertically integrated manufacturing environment.
- Design sealing fixtures, tooling, and equipment in SolidWorks.
- Create and maintain process documentation and work instructions; collaborate with manufacturing, quality, and design teams.

Machine Learning Engineer – EEG & Biosignal Processing San Francisco, CA
University of California, San Francisco Health | Amorim Lab Aug 2025 – May 2026

- Developed and validated multi-stage EEG and biosignal processing pipelines for a large-scale clinical dataset; performed root cause analysis on output issues and iterated to resolution.
- Prepared technical reports summarizing methods, results, and recommendations; presented findings to cross-functional clinical and engineering collaborators.

Student Researcher – Microfluidics & Biological Systems Long Beach, CA
Autonomous Machinery for Engineering Living Matter Lab, CSULB Apr 2024 – May 2025

- Designed and fabricated microfluidic device prototypes from concept through testing; iterated on designs based on experimental results and troubleshot fabrication issues.

Tutor – Biomedical Engineering Courses Long Beach, CA
California State University, Long Beach Jan 2025 – May 2025

- Tutored students across biomedical engineering coursework, including upper-division courses.

Teaching Assistant – Biomedical Signal Processing Long Beach, CA
California State University, Long Beach Aug 2024 – Dec 2024

- Developed instructional materials and guided 30+ students through hands-on lab work in instrumentation and data analysis; communicated technical concepts clearly to varied backgrounds.
- Reinforced core concepts for students through lab guidance and one-on-one support.

PROJECTS

EEG-Based Neurological Outcome Prediction (MEng Capstone) May 2026

- Built a multi-stage EEG analysis pipeline achieving 93% AUC-ROC in predicting neurological recovery across a dataset of ~1,000 coma patients; reported results to clinical and engineering stakeholders.

VR-Integrated Low-Cost EEG Brain-Computer Interface (Undergraduate Capstone) May 2025

- Analyzed EEG recordings from a team brain-computer interface study, applying filtering and frequency analysis to show that a VR display evoked significantly stronger brain responses than a laptop screen.

SKILLS

Signal Processing & Analysis: EEG/biosignal processing, machine learning, statistical analysis, BioRadio, MATLAB, Python
Teaching: Lab instruction, tutoring, instructional materials, explaining technical concepts to varied audiences
Design & Fabrication: SolidWorks, AutoCAD, GD&T, engineering drawings, microfluidic fabrication, laser cutting, test setup design
Documentation & Professional: Technical reports, SOP development, cross-functional collaboration, Excel, Word, PowerPoint